

WHO guidelines for the clinical management of filovirus disease

Executive summary

World Health Organization

WHO guidelines for the clinical management of filovirus disease

Executive summary

World Health Organization

© World Health Organization 2026

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: "This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition".

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization (http://www.wipo.int/amc/en/mediation/rules/).

Suggested citation. WHO guidelines for the clinical management of filovirus disease: executive summary. Geneva: World Health Organization; 2026. https://doi.org/10.2471/B09786. Licence: CC BY-NC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at http://iris.who.int.

Sales, rights and licensing. To purchase WHO publications, see https://www.who.int/publications/book-orders. To submit requests for commercial use and queries on rights and licensing, see https://www.who.int/copyright.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

Design and layout by L'IV Com Sàrl.

Executive summary

Background

This is a summary of the *WHO guidelines on the clinical management of filovirus disease*. The full text, which includes all of the references to source data and publications is available at https://doi.org/10.2471/B09774. The whole guideline is also available in interactive format online at https://app.magicapp.org/#/guideline/11037.

Abbreviations

aOR	adjusted odds ratio
CRT	capillary refill time
GDG	guideline review committee
IPC	infection prevention and control
OR	odds ratio
PCR	polymerase chain reaction
RCT	randomized controlled trial
RNA	ribonucleic acid
RR	relative risk
SAE	severe adverse event
WHO	World Health Organization

Clinical question: Which interventions can improve patient outcomes from filovirus disease?

Rationale and scope

Filovirus disease includes infections caused by Bundibugyo, Ebola, Marburg, Sudan and Taï Forest viruses. Previous WHO clinical guidance focused on Ebola virus disease, and was based on expert-defined best clinical practice. These guidelines expand the scope to all filoviruses to reduce duplication, harmonize approaches and maximize impact across outbreaks. Recommendations have also been formulated using data from systematic evidence searches, and according to the GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework to maximize transparency and to ensure implementation considerations have been carefully evaluated.

Methods

The guideline was developed in accordance with the *WHO handbook for guideline development*. A multidisciplinary, geographically representative Guideline Development Group (GDG) was convened, with representation across clinical disciplines, regions and genders. All members completed declarations of interest, which were reviewed and managed using standard WHO procedures.

Clinical questions were framed using the PICO (Population, Intervention, Comparison/Comparator, Outcomes) format. Evidence synthesis relied primarily on recent high quality systematic reviews. The GRADE approach was used to assess certainty of evidence and to determine the direction and strength of recommendations. For each recommendation, an evidence-to-decision framework was applied, incorporating: balance of benefits and harms; certainty of evidence; patient values and preferences; feasibility; resource use; equity; acceptability.

Draft recommendations underwent internal and external peer review and final approval by the WHO Guideline Review Committee.

Key recommendations

- Systematic patient monitoring should be established for patients with suspected or confirmed filovirus disease, using structured and documented clinical assessment at regular intervals, with frequency adapted to patient condition (Good practice statement).

- WHO recommends that for patients with suspected or confirmed filovirus disease, systematic laboratory monitoring be performed based on clinical assessment, rather than on routine testing (Strong recommendation for, low certainty of evidence).

- WHO recommends patients with suspected or confirmed filovirus disease with ongoing gastrointestinal fluid losses and some dehydration, be treated with a protocolized oral rehydration regimen (e.g. WHO Plan B) (Strong recommendation for, moderate certainty evidence).

- WHO suggests patients with suspected or confirmed filovirus disease with ongoing gastrointestinal fluid losses and severe dehydration be treated with a protocolized intravenous fluid regimen compared with no protocolized intravenous fluid regimen (Conditional recommendation for, low certainty evidence).

- WHO recommends balanced crystalloids (such as Ringer's lactate) rather than 0.9% sodium chloride for acute intravenous fluid resuscitation in patients with suspected or confirmed filovirus disease (Strong recommendation for, moderate certainty of evidence).

- WHO suggests for patients with suspected or confirmed filovirus disease and clinical signs of shock or circulatory impairment, the serial measurement of lactate as part of the assessment of perfusion to guide further intravenous fluid management (Conditional recommendation for, low certainty of evidence).

- WHO suggests for patients with suspected or confirmed filovirus disease and clinical signs of shock or circulatory impairment, capillary refill time should be used as part of the assessment of perfusion to guide intravenous fluid management (Conditional recommendation for, low certainty of evidence).

- WHO suggests that patients with suspected or confirmed filovirus disease with shock who are being resuscitated with intravenous fluids, vasopressor treatment be initiated early rather than later (Conditional recommendation for, low certainty of evidence).

- WHO recommends that patients with suspected or confirmed filovirus disease requiring a vasopressor be treated with norepinephrine rather than dopamine (Strong recommendation for, moderate certainty evidence).

- WHO suggests that patients with suspected or confirmed filovirus disease requiring a vasopressor be treated with norepinephrine rather than epinephrine (Conditional recommendation for, very low certainty of evidence).

- WHO suggests that for patients with filovirus disease requiring initiation of vasopressors, peripheral intravenous catheters be used rather than central venous catheters (Conditional recommendation for, very low certainty of evidence)

- WHO recommends that for patients with suspected or confirmed filovirus requiring central venous access, ultrasound guided placement be used rather than non-ultrasound guided placement (Strong recommendation for, moderate certainty of evidence).

- WHO suggests for the use of presumptive broad-spectrum antibiotics in patients with confirmed filovirus disease in the presence of suspected sepsis or bacterial co-infection (Conditional recommendation for, low certainty of evidence).

- WHO suggests against the use of tranexamic acid in patients with confirmed filovirus disease and haemorrhage (Conditional recommendation against, very low certainty of evidence).

- WHO suggests for the use of tranexamic acid in patients with confirmed filovirus disease and postpartum haemorrhage (Conditional recommendation for, low certainty evidence).

- WHO recommends that persons who have survived/recovered from filovirus disease be offered structured follow-up after discharge (Strong recommendation for, moderate certainty of evidence).

Practical applications

High-quality supportive care for filovirus disease can reduce mortality. Implementation of clinical interventions is complicated by resource availability, the suboptimal environment in which care is often delivered, the need for stringent infection prevention and control (IPC) procedures and limited staff training. This clinical guideline complements multiple other WHO guidance documents, which together address these challenges. As part of this multidisciplinary approach, applying clinical principles and delivering individualized person-centred care will save lives and improve the outcomes for patients.

Good practice statement

Systematic patient monitoring should be established for patients with suspected or confirmed filovirus disease, using structured and documented clinical assessment at regular intervals, with frequency adapted to patient condition.

- This statement applies to all children and adults, including pregnant women.
- Abnormalities which are identified during an assessment and monitoring must be followed by immediate action which is expected to improve the patient's outcome.

The panel unanimously emphasized the importance of monitoring, and that this be linked directly to clinical interventions, with individualized decision-making to ensure patients' needs are met. The panel postulated very low frequency adverse events related to non-invasive monitoring, and trivial impact where these occurred. By contrast, the panel considered a potential for catastrophic harm when abnormal physiology is not recognized.

The panel were initially presented evidence from randomized controlled trials (RCTs) of continuous telemonitoring, which found very low certainty evidence for outcomes of mortality, major complications and length of stay. However, they felt these did not to reflect the real risk:benefit of broader patient monitoring, as this modality was not the expected method of delivering frequent, intermittent monitoring in filovirus outbreaks.

The lack of RCTs was considered to be due to lack of equipoise due to the fundamental principle of monitoring being applied throughout clinical practice. The panel discussed the possibility of creating a linked evidence profile to support a non-discordant strong recommendation. However, this would require disproportionate effort. Experiential evidence was shared about how early detection of abnormalities if acted upon does save lives, and there was universal agreement that patient observations are a fundamental part of delivery of high-quality medical care.

Supporting this were considerations of equity (all patients benefit) and demonstrated acceptability and feasibility as part of everyday practice. Resource needs were discussed including sufficient staff and training. The absolute human resource requirement would be determined by the level of care demanded in the treatment centre.

The panel advocated for improved treatment centre design to facilitate monitoring from the green zone, and communication between the green and red zones.

The panel underlined that the resulting good practice statement had the same strength as a strong recommendation in favour of systematic patient monitoring. The panel underlined the important principle that frequency of monitoring be based on disease severity.

Strong recommendation for, low certainty of evidence

WHO recommends that for patients with suspected or confirmed filovirus disease, systematic laboratory monitoring be performed based on clinical assessment, rather than on routine testing.

- This recommendation applies to all children and adults, including pregnant women.
- Laboratory monitoring should be linked to actionable clinical interventions intended to improve patient outcomes. The laboratory result turnaround time should be appropriate to inform clinical decision-making.
- The scope and frequency of laboratory testing should be tailored to the needs of the patient, with selection to use resources wisely and minimize venous punctures.
- The minimal set of available tests should include: point-of-care malaria rapid diagnostic tests, glucose, haemoglobin/haematocrit, complete blood count, pregnancy test, urinary dipstick, tests required for blood transfusion, electrolytes and biochemistry. Additional tests may be required based on level of care.
- Biosafety must be implemented during all phases, including sample collection, transport, laboratory manipulation and sample disposal.

Evidence was derived from a systematic review commissioned by WHO, published as part of these guidelines.

The panel reasoned that laboratory monitoring is a tool which enables earlier recognition of important pathophysiologic abnormality, and provides key data related to disease. Through directly informing clinical action, laboratory results can improve patient outcomes. The panel gave examples of hypoglycaemia, which is life threatening, and hyperkalaemia, which is both life threatening and impossible to detect without laboratory testing. In both cases, specific treatment can correct the abnormality, thereby reducing the risk of death. Other examples included the detection of kidney or liver impairment for which clinical adjustments in drug dosing are required to avoid iatrogenic harm.

Univariate analysis demonstrated a clear association between abnormality and in-facility mortality: for hyperkalaemia OR 11.11 (95% CI 2.70–45.66), moderate certainty; for rhabdomyolysis aOR 2.18 (95% CI 1.09–4.37); for creatinine aOR 2.3 for each 100 mmol/L increase (95% CI 1.1–4.6), high certainty; and for haemoglobin aOR 0.67 (95% CI 0.47–0.93), high certainty.

Using this linked evidence, overall judged to be low certainty, the panel made a strong recommendation for laboratory monitoring. The panel noted that many life-threatening biochemical abnormalities had no good clinical correlation, but could be treated effectively when identified. A discordant recommendation was therefore justified on the basis of the high potential for catastrophic outcome without the ability to perform laboratory testing.

Experience from the most recent filovirus disease outbreaks demonstrated that laboratory monitoring was not "standard of care", and may in some circumstances be aspirational leading to delayed recognition and adverse patient outcomes. The panel stated that a strong recommendation would be more likely to drive improvements in availability by sending a clear message to policy-makers, public health laboratory administrators and health facility administrators.

The panel listed a core set of laboratory tests that should be made available, and that additional tests be made available based on level of care provided in the treatment centre.

Core tests included: sodium (Na+); potassium (K+); lactate; creatinine; glucose; complete blood count (red blood cells, white blood cells, haemoglobin, haematocrit, platelets); prothrombin time/activated partial thromboplastin time (PT/APTT); malaria rapid diagnostic test (MRDT); HIV as per national protocol; human chorionic gonadotropin (bHCG) for pregnancy test, blood group and matching (for transfusion purposes); urinalysis (urine dipstick).

Comprehensive tests included core tests plus: urea; chloride; bicarbonate; calcium; phosphate; magnesium; creatinine kinase; liver transaminases; fibrinogen; platelets; blood culture; multiplex PCR pathogen testing; c-reative protein (CRP). In addition, in locations with high HIV prevalence, consideration of urinary lipoarabinomannan (LAM) and cryptococcal antigen for patients living with HIV.

Strong recommendation for, moderate certainty of evidence

WHO recommends patients with suspected or confirmed filovirus disease with ongoing gastrointestinal fluid losses and some dehydration[a], be treated with a protocolized oral rehydration regimen (e.g. WHO Plan B).

- This recommendation applies to both children and adults, including pregnant women.
- Systematic monitoring of vital signs including fluid input and output is required to identify those patients that may need escalated treatment with intravenous fluid.
- Peripheral venous catheter should be placed in all patients with filovirus disease in case there is need to augment with intravenous fluids.
- This does not apply to patients with sepsis or septic shock, for which intravenous fluid therapy is indicated, nor those with severe acute malnutrition, for which other WHO guidelines should be used.

[a] **In children:** Some dehydration is defined as no signs of severe dehydration, with two or more of the following signs: restlessness; irritability; sunken eyes; thirsty and drinks eagerly; skin pinch goes back slowly. Severe dehydration is defined as two or more of the following signs: lethargy or unconsciousness; sunken eyes; unable to drink or drinks poorly; skin pinch goes back very slowly (≥ 2 seconds).

In adults: Some dehydration is less well defined than for children, but includes symptoms of: feeling thirsty; concentrated urine; infrequent urination; feeling dizzy or lightheaded; tiredness; dry mouth, lips and tongue; sunken eyes.

Evidence was drawn from large observational cohorts from cholera, taken before and after the implementation of structured oral fluids (now codified as WHO Plan B). The certainty of evidence was upgraded due to large effect size. The panel discussed the applicability to filovirus disease, and noted that where some dehydration was present, the benefits were likely to be similar. The panel emphasized that most patients with some dehydration would place high value on moderate certainty evidence of mortality impact with an oral rehydration regimen (i.e. WHO Plan B), especially when the harms are insignificant relative to the mortality benefits. The panel recognized that clinically important differences would increase in patients with severe dehydration and/or sepsis and/or shock, or those at high risk of complications from high-volume fluid therapy such as those with pre-existing congestive heart failure, or severe acute malnutrition.

The panel emphasized that this recommendation was only applicable where patients could take oral fluids easily. They noted that for patients with filovirus disease who have the potential for rapid clinical deterioration, intravenous access should be placed even where oral fluids are being given, and close monitoring is required. Similarly, the panel stressed that the presence of sepsis and/or shock necessitates intravenous fluid therapy.

Conditional recommendation for, low certainty evidence

WHO suggests patients with suspected or confirmed filovirus disease with ongoing gastrointestinal fluid losses and severe dehydration[a] be treated with a protocolized intravenous fluid regimen compared with no protocolized intravenous fluid regimen.

- This recommendation applies to both adults and children.
- Systematic and frequent monitoring of haemodynamic status, fluid input and output, and signs of fluid overload is required to titrate and individualize fluid therapy.
- Modifications of WHO Plan C are required to enhance monitoring to enable more restrictive fluid regimen in cases where it is difficult to discriminate between severe dehydration and sepsis-induced hypoperfusion (see Practical info section).
- Once dehydration is corrected, administer maintenance fluids based on calculated requirements based on weight and insensible and ongoing losses.
- Patients that develop shock, need careful re-evaluation for etiology of shock to guide appropriate resuscitation.

[a] **In children:** Severe dehydration is defined as two or more of the following signs: lethargy or unconsciousness; sunken eyes; unable to drink or drinks poorly; skin pinch goes back very slowly (≥ 2 seconds).

In adults: Severe dehydration, which requires urgent resuscitation with intravenous fluid, is suggested by any of: systolic blood pressure is less than 100 mmHg; heart rate is more than 90 beats per minute; capillary refill time is more than 2 seconds; peripheries are cold to touch; respiratory rate is more than 20 breaths per minute.

The panel emphasized that majority of patients with severe dehydration and ongoing gastrointestinal losses would place high value on low certainty evidence of mortality impact with an intravenous rehydration regimen (such as Plan C) and expressed strong concern about the potential serious harms when clinical distinctions between sepsis and severe dehydration are evident in these recommendations. The panel noted that the use of Plan C was widely known by health and care providers treating patients with severe dehydration from other diarrhoeal causes and thus change of practice would require a stronger evidence base. However, adding modifications to the intravenous hydration regimen (modified Plan C, see Practical info) to mitigate harm and optimize effectiveness in populations where risk of harm may be greater such as patients with filovirus disease and sepsis and those with other chronic disease predisposed to volume overload (renal failure, liver failure, cardiac insufficiency, etc.).

This applies to adults, children and pregnant women. However, the panel noted that for children with coexisting severe acute malnutrition, other fluid protocols should be considered. For patients with filovirus disease and sepsis, see subsequent recommendations.

Strong recommendation for, moderate certainty of evidence

WHO recommends balanced crystalloids (such as Ringer's lactate) rather than 0.9% sodium chloride for acute intravenous fluid resuscitation in patients with suspected or confirmed filovirus disease.

- This applies to children and adults including pregnant women, who have sepsis, septic shock and/or dehydration.
- Conditions where 0.9% sodium chloride may be preferred include patients with filovirus disease that have trauma including traumatic brain injury.

The panel felt that the majority of patients would prefer treatment with balanced solution over normal saline, placing high value on the moderate certainty evidence of mortality benefit (using the 600/1000 baseline mortality risk, equating to 42 fewer per 1000, 95% CI 84 fewer to 6 more). In terms of harms, the panel discussed increased prevalence of hyperchloraemic acidosis with 0.9% sodium chloride as a potential mechanism. However, the panel also noted the possible increase in harm if balanced solutions were used in traumatic brain injury. The panel also discussed that monitoring electrolytes in filovirus patients with vomiting and diarrhoea may also impact choice of intravenous fluid. Consensus was achieved, but four panel members stated abstentions which related to specific circumstances in which Ringer's lactate may not be superior to 0.9% sodium chloride. These have been noted in the recommendation remarks, but were expected to be rare in the primary presentation of filovirus disease and thus did not require a conditional recommendation.

Given the lack of sepsis-specific data on renal replacement therapy and ventilator-free days in the meta-analysis, the panel considered data from critically unwell patients more broadly. Additional indirectness in these endpoints was considered and resulted in rating down once, potentially mediated through differences in population (the more general critically unwell study populations included those with traumatic brain injury or advanced liver disease rather than sepsis only).

The panel judged that the recommendation would apply similarly in children, pregnancy and during breastfeeding. The concordance with WHO paediatric fluid guidelines was noted. This recommendation does not apply to patients with filovirus disease and trauma and traumatic brain injury.

The panel noted the cluster randomized trial that was recently published and not included in the systematic review that informed this recommendation. This study evaluated the use of balanced solution versus normal saline at hospital level and did not show difference on primary outcome of mortality and was thus deemed too indirect to include in the deliberations.

Conditional recommendation for, low certainty evidence

WHO suggests for patients with suspected or confirmed filovirus disease and clinical signs of shock or circulatory impairment, the serial measurement of lactate as part of the assessment of perfusion to guide further intravenous fluid management.

- This recommendation applies to children and adults, including pregnant women.
- Lactate is a marker of poor perfusion and when elevated is associated with poor prognosis.
- The assessment of perfusion also includes other clinical markers such as blood pressure, heart rate, urine output, mental status and capillary refill time.
- Reduction of lactate over time can be used to infer the effectiveness of fluid replacement. However, normalization of lactate should not be the target.
- There are situations where elevated lactate levels are not associated with poor perfusion and these include: advanced liver disease when lactate metabolism is impaired, excessive muscles activity, focal or regional areas of tissue ischaemia, and some drugs (for example metformin). Persistent raised lactate should prompt a search for other coexistent pathologies.
- Lactate can be measured in arterial, venous or capillary blood. The choice of sample type should follow local practice and be used consistently, especially when comparing lactate levels over time. Arterial puncture is the reference standard for accuracy, but it is not encouraged due to bleeding risk in filovirus disease.

The panel chose a conditional recommendation, judging that the majority of patients, would choose to have lactate added to assessment of perfusion as it may reduce mortality. However, this evidence was not compelling enough to warrant universal use where resource limitations were significant. The panel also noted that lactate is not used alone but rather as part of an assessment of perfusion, where serial measurements of lactate are used to guide fluid management and that normalization of lactate should not be a target. In the included RCTs, there were variable standards of care, and many used mean arterial pressure, central venous pressure monitoring and other tools to monitor perfusion. The panel noted that there was uncertainty how this would apply in situations with more limited monitoring as "standard of care". A strong emphasis was placed on instituting the best possible monitoring given local resources, using all the measures available, including basics such as blood pressure, heart rate, urine output and mental status and capillary refill time (CRT) (see next recommendation). The panel also raised concern about use of lactate in filovirus disease patients with liver dysfunction and should be considered as an area for further research.

In addition, lactate could be used for identifying patients at high risk of adverse outcome. The panel judged that the recommendation should apply to adults and children and pregnant women given similar associations with outcome, and the use of lactate as part of frequently used sepsis definitions and screening tools.

RCTs used mostly arterial measurements as these are considered most accurate, although other modalities were allowed. In field implementation, venous or capillary measures might be more frequently used. The panel considered that while this might introduce some indirectness, this was unlikely to be an important consideration especially where the lactate was used to consider dynamic changes in individual patients.

Conditional recommendation for, low certainty evidence

WHO suggests for patients with suspected or confirmed filovirus disease and clinical signs of shock or circulatory impairment, capillary refill time should be used as part of the assessment of perfusion to guide intravenous fluid management.

- This recommendation applies to children and adults, including pregnant women.
- The assessment of perfusion also includes other markers such as blood pressure, heart rate, urine output, mental status and lactate.
- Use of capillary refill time can also be integrated into assessments using complementary technologies such as ultrasound, depending on availability, training and level of care.
- Use of standardized protocols for capillary refill time assessment will reduce variability in measurements and increase reliability and accuracy.

The panel considered the ANDROMEDA-SHOCK2 study as the primary source of evidence. The addition of data from a second RCT of 30 patients was felt to be uninformative. The evidence from patients with septic shock was judged sufficiently direct to filovirus disease not to reduce level of certainty. The panel considered the data for separate endpoints (mortality, vital support-free days and hospital length of stay) although these were secondary endpoints within the trial. The panel noted that the primary (hierarchical) endpoint from the trial gave low certainty evidence of superiority of capillary refill-driven intravenous fluid treatment (win ratio 48.9% vs 42.1%), and that this was mostly driven by a difference in vital support-free days. In resource-limited settings, the panel judged that reduction in ventilation particularly, captured within the vital support-free days, would represent an important benefit. Harms were not directed captured, although would be expected to be represented in the outcomes above, and the measurement itself has no important associated harms.

Given the simple and low-cost nature of the test, and universal ability to perform, the panel felt that a conditional recommendation for CRT was appropriate.

The panel noted that when implemented in clinical practice, CRT is not used alone, and rather should be used within a protocolized resuscitation which includes assessment of perfusion through blood pressure, heart rate, urine output and lactate, where available. It was suggested that the strength of pulse oximetry trace might also be incorporated by clinicians into a overall impression of circulatory sufficiency. Members noted that CRT is more established in the assessment of children but its use in adults was less common so standardization of measurement would be important (see Practical info).

The panel judged that the recommendation should apply to all ages in view of RCT evidence from adults which was felt to be relevant to children, and its incorporation as standard of care within paediatric clinical practice.

Conditional recommendation for, low certainty evidence

WHO suggests that patients with suspected or confirmed filovirus disease with shock who are being resuscitated with intravenous fluids, vasopressor treatment be initiated early rather than later.

- This recommendation applies to children and adults, including pregnant women.
- Vasopressors should be used within a resuscitation strategy that includes: appropriate intravenous fluid therapy; frequent monitoring, including for cardiac arrhythmia.
- Individualize titration of the dose to desired effect (targeting arterial pressure as an indicator of perfusion).
- To minimize adverse effects, use the lowest dose possible for desired effect and discontinue vasopressors when no longer indicated.

The recommendation does not apply to shock caused predominantly by severe dehydration with ongoing gastrointestinal losses, where fluid replacement should be the primary treatment for circulatory insufficiency.

Strong recommendation for, moderate certainty of evidence

WHO recommends that patients with suspected or confirmed filovirus disease requiring a vasopressor be treated with norepinephrine rather than dopamine.

- This recommendation applies to children and adults, including pregnant women.
- Vasopressors should be used within a resuscitation strategy that includes: appropriate intravenous fluid therapy; frequent monitoring, including for cardiac arrhythmia.
- Individualize titration of the dose to desired effect (targeting arterial pressure as an indicator of perfusion).
- To minimize adverse effects, use the lowest dose possible for desired effect and discontinue vasopressors when no longer indicated.

The panel's strong recommendation for norepinephrine was driven by evidence of mortality reduction as a critically important outcome, and supported by lower rates of observed arrhythmias and serious adverse events. The evidence from sepsis trials was felt to be sufficiently direct to not rate down the certainty of evidence. The panel weighed the arrhythmia outcome in their decision as although it was not previously prioritized, it was judged an important and proximal marker of safety/harm.

No direct RCT comparisons were available for paediatric populations. The panel judged that this recommendation was appropriate to children and adults, considering the current clinical use of norepinephrine or epinephrine as first choice vasopressor in recent trials of early therapy.

While there were no direct data on the use in pregnancy, the panel judged that the clear mortality benefit in favour of norepinephrine would likely also apply to pregnant women. This is consistent with principles of obstetric practice in critical illness; optimizing maternal circulation is the most effective strategy to preserve fetal perfusion, and delays in vasopressor initiation may worsen outcomes for both mother and fetus.

Conditional recommendation for, low certainty evidence

WHO suggests that patients with suspected or confirmed filovirus disease requiring a vasopressor be treated with norepinephrine rather than epinephrine.

- This recommendation applies to children and adults, including pregnant women.
- The choice of norepinephrine or epinephrine may depend on availability and local practices.
- For some patients, for example, those with cardiac systolic dysfunction, epinephrine may be preferable to norepinephrine alone.
- Vasopressors should be used within a resuscitation strategy that includes appropriate fluid therapy, frequent monitoring, including early identification of arrythmia.
- Individualize titration of the dose to desired effect (improved perfusion).
- To minimize adverse effects, use the lowest dose possible for desired effect and discontinue vasopressors when no longer indicated.

The panel judged that a conditional recommendation for norepinephrine over epinephrine would ensure more consistent use of the vasopressor with the most evidence on efficacy, while leaving room for clinicians to use epinephrine in settings where norepinephrine is not available or where cardiac insufficiency may warrant epinephrine preferentially.

The panel weighed the arrhythmia outcome in their decision and though it was not previously prioritized, it was judged an important and proximal marker of safety/harm. Estimates of absolute risk for mortality were used from the original meta-analysis and were not corrected for different baseline risks given the very wide confidence intervals.

Theoretical considerations of harm were considered with use of epinephrine and its higher β-1 antagonism leading to enhanced arrhythmogenicity and splanchnic vasoconstriction. The panel also noted its previous strong recommendation for use of norepinephrine over dopamine based on moderate certainty evidence for mortality and this would increase its availability and use in general. The panel noted that trials which included norepinephrine as a comparator also included patients who received concurrent inotropic agents. However, there was no subgroup interaction effect for norepinephrine alone compared with norepinephrine with another inotropic (e.g. dopexamine), $p = 0.56$. These data were consistent with a network meta-analysis which estimated the odds ratio for mortality of norepinephrine (+/- dobutamine) versus epinephrine of 0.87 (95% CI 0.67–1.12).

The GDG judged that this recommendation should apply to adults and children. This was supported by data from an RCT of 67 children which suggested no difference in mortality between norepinephrine (+dobutamine) and epinephrine treatment (RR 0.59; 95% CI 0.28–1.25), but which was underpowered to confidently inform on benefits or harms.

While there were no direct data on the use in pregnancy, the panel judged there was no reason to believe that decisions on epinephrine versus dopamine in pregnant women with a life-threatening infection would have a different harm:benefit balance than the wider population, which is supported by current clinical practice.

Conditional recommendation for, very low certainty of evidence

WHO suggests that for patients with filovirus disease requiring initiation of vasopressors, peripheral intravenous catheters be used rather than central venous catheters.

- This recommendation applies to children and adults, including pregnant women.
- Peripheral intravenous administration avoids the delay associated with central venous catheterization and enables more timely treatment which may improve outcomes.
- Use the largest bore peripheral intravenous catheter possible to minimize the risk of extravasation.
- Systematic and frequent monitoring of intravenous site is required.
- Transition to the use of central venous administration should be anticipated.

Evidence was derived from a systematic review commissioned by the WHO, published as part of these guidelines.

The panel made a conditional recommendation on the basis of their field experience of the time taken to initiate treatment being shorter for peripheral administration, and in conjunction with evidence that earlier vasopressor treatment was likely to improve important outcomes (see early vasopressor recommendation).

The panel judged that initiation of vasopressors via peripheral rather than central venous catheters would be easier, faster and lead to more rapid initiation of life-saving medicine (see norepinephrine recommendation). Concerns were raised about the risk of extravasation with peripheral administration, which requires extremely careful monitoring and immediate clinical action to prevent serious harm. Meta-analysis of observational cohorts suggest this occurs at rates of 14 per 1000 (95% CI 7.2–23.2) based on 93 events in 4613 participants from 19 studies. The panel emphasized that peripheral administration would be a bridge to necessary placement of a central venous catheter.

Strong recommendation for, moderate certainty of evidence

WHO recommends that for patients with suspected or confirmed filovirus requiring central venous access, ultrasound guided placement be used rather than non-ultrasound guided placement.

- This recommendation applies to children and adults, including pregnant women.
- Ultrasound guided central venous catheter placement requires trained health workers, essential equipment and supplies.
- Ensure appropriate IPC measures are in place to reduce risk of infection when using ultrasound.
- Systematically monitor the catheter site.
- Remove the catheter when no longer needed.

The panel felt that the safety of patients could be impacted by their other recommendations for vasopressor use. Particularly that, while initial peripheral administration was preferred, the use of central venous catheterization would be more common if all the recommendations were enacted. The panel therefore considered the evidence for the mode of placement of central venous catheter, which might mitigate risks of placement of catheters.

Data were drawn from a systematic review of ultrasound vs landmark approaches for jugular vein cannulation which included multiple reasons for insertion. The evidence was considered to directly pertain to filovirus disease patients as the procedure's risks and benefits were expected to be similar. The panel noted that bleeding in the context of filovirus-related haemorrhagic syndrome might be expected to worsen outcomes related to arterial puncture. Although outcomes had not been prioritized, the GDG had pre-agreed that serious adverse events (SAE) was a critical outcome. In the context of venous cannulation, SAE included arterial puncture, pneumothorax and severe subcutaneous emphysema.

The panel unanimously agreed that a recommendation for ultrasound-guided placement be made. A minority, citing feasibility concerns, felt that this should be a conditional recommendation despite moderate certainty evidence of benefit. However, after deliberation, a vote was taken in which 80% of the panel opted for a strong recommendation. The justification for this was the high risk of catastrophic outcomes if bleeding or pneumothorax occurred in the context of patients with increased bleeding risk, and within a complex environment constrained by resources and the need for procedures performed with full personal protective equipment (PPE). The panel felt that ultrasound and associated materials for IPC were essential, and noted that skilled workforce and safety protocols were also mandatory.

Conditional recommendation for, low certainty evidence

WHO suggests for the use of presumptive broad-spectrum antibiotics in patients with confirmed filovirus disease in the presence of suspected sepsis or bacterial co-infection.

- This recommendation applies to children and adults, including pregnant women.
- Presumptive antibiotics means use when there is clinical suspicion of sepsis and/or bacterial co-infection.
- Administration of intravenous antibiotics should occur without delay, where there is a suspicion of sepsis, or shock.
- When possible, antibiotic choice should be based on local antimicrobial resistance (AMR) patterns.
- This does not apply to patients with low likelihood of bacterial co-infection. These patients should not be routinely given antibiotics, but rather monitored and re-assessed for the presence of bacterial infection and sepsis to guide antibiotic treatment.
- Use available laboratory tools, including bacteriology and biomarkers, to guide microbiological diagnosis and antimicrobial therapy with appropriate de-escalation when possible.

Evidence was derived from a systematic review commissioned by the WHO, published as part of these guidelines.

The panel determined that in patients with confirmed filovirus disease that presumptive antibiotics (the administration without objective clinical evidence of bacterial infection) may reduce mortality. They reasoned that the major clinical manifestation of severe filovirus disease is a sepsis-like syndrome, and bacterial co-infection cannot be excluded. There is also a postulated increased translocation of bacteria from the gastrointestinal tract, as is seen in other critical illness. Given the pre-existing high mortality rate, in a population with severe disease, the large expected benefit from antibiotics where undiagnosed bacterial infection takes place, the panel felt that the individual patient benefit was likely to outweigh potential public harms associated with AMR, and the individual concerns of adverse events.

The panel qualified that AMR was a global threat, but the number of patients to whom this recommendation would be applied would be very small, and their contribution as a driver of AMR would be negligible. However, good practice would determine that the most appropriate agent should be used for the shortest time possible.

For non-severe cases of filovirus disease, and where patients have no evidence of bacterial co-infection, the panel questioned the benefit and noted that the possible harms related to severe adverse events and AMR remained. This recommendation should not be applied under these circumstances. The panel noted that there was no formal tool which established a pre-test likelihood of bacterial infection, and that the physician should decide on the basis of an individualized risk assessment.

The panel judged that the same justification would apply to children and adults, including pregnant women.

The recommendation applies at the point of admission to the isolation unit for suspected cases and/or filovirus treatment centre. The possibility of subsequent hospital acquired infection, with later onset and alternative bacterial etiologies necessitates an individualized approach. The panel noted that determining etiology through bacterial culture and antibiogram would be best practice outside of outbreaks, but rarely available to filovirus treatment centres.

Conditional recommendation against, very low certainty of evidence

WHO suggests against the use of tranexamic acid in patients with confirmed filovirus disease and haemorrhage.

- This recommendation applies to adults and children with filovirus disease that have bleeding from intravenous sites, gastrointestinal tract and oral mucosa.
- This does not apply to patients with filovirus that are bleeding due to a condition for which tranexamic acid has a proven benefit, such as postpartum haemorrhage, surgical bleeding or trauma.

There is a clear need for more biological studies to understand the pathophysiological mechanisms of bleeding in filovirus disease. The panel agreed that the current mechanistic understanding from sepsis, where disseminated intravascular coagulation (DIC) is an important contributor to altered haemostasis, raises strong concerns about potential harm associated with tranexamic acid. The closest parallel source of evidence was judged to be from delayed treatment of patients after trauma, because DIC is likely to be a common pathophysiological process. Here, tranexamic acid is associated with increased mortality.

Additional concern was raised by the low certainty evidence of increased thrombosis in patients with gastrointestinal haemorrhage. These suggestions of harm, in the absence of direct evidence of benefit led the panel to make a conditional recommendation against tranexamic acid.

This recommendation applies to children and adults, and pregnant women who do not have postpartum haemorrhage; the panel decided to make a separate recommendation for the use in postpartum haemorrhage.

Conditional recommendation for, low certainty evidence

WHO suggests for the use of tranexamic acid in patients with confirmed filovirus disease and postpartum haemorrhage.

- Employ standard care protocols for postpartum haemorrhage, including a uterotonic (oxytocin is preferred where available).
- Tranexamic acid should be given intravenously within 3 hours of birth.

Given the proven benefit in postpartum haemorrhage, the panel judged that implementation of tranexamic acid treatment in this specific subpopulation would be likely to be beneficial. While the potential for theoretical risk of harm related to DIC remained, this was not quantifiable and the availability of broader evidence for benefit in postpartum haemorrhage was robust. The panel judged that treatment should therefore be instituted in line with WHO recommendations. For those with filovirus disease, the panel felt there was indirectness of evidence from the general population, which resulted in reduced certainty (rated down to low), and a subsequent conditional recommendation.

Strong recommendation for, moderate certainty of evidence

WHO recommends that persons who have survived/recovered from filovirus disease be offered structured follow-up after discharge.

- This recommendation applies to children and adults.
- Structured follow-up includes access to free and comprehensive care to assess and treat complications such as mental health symptoms, ocular disease, and persistence of virus in sanctuary sites; as well as ensure safe delivery for pregnant women who have recovered.

The panel originally considered evidence which included the prevalence of symptoms and medical abnormalities after filovirus infection. However, without evidence of interventions which would be expected to have a benefit, the evidence was considered of low certainty. Further work was therefore undertaken to identify multiple areas of linked evidence with the potential for significant improvement in patient-important outcomes. The panel agreed that the complexity of care required, together with high rates of post-infection chronic symptoms would be unlikely to be adequately treated in existing health systems without specific guidance. The interventions would be most efficiently delivered as a package of assessment and treatment at post-filovirus follow-up clinics, which could integrated into primary health services and potentially lead to local strengthening of capacity.

The following were identified in terms of evidence linking the incidence of disease with positive outcomes from its treatment:

1. Screening for depression, with subsequent initiation of selective serotonin reuptake inhibitors (SSRIs), which have been shown to improve patient-important mental health outcomes. Depression was considered a critical outcome as "avoiding stigma and mental health issues" had a median score 8/9 on a prioritization exercise. The panel estimated that a maximum of 25% of those 150 per 1000 survivors with depression would receive treatment (i.e. would have depressive symptoms recognized and treated if typical usual care approaches were taken, i.e. reactive, patient-driven care-seeking). This is consistent with WHO reported figures for the proportion of people with severe mental disorders who receive no treatment for their disorder in LMICs of 76–85%. Using the conservative estimate (76%), the number of expected additional cases of treated depression if structured follow-up were employed would be 16/1000, and would exceed the minimal important difference of 15/1000.
2. Reduction of onward transmission related to sexual activity through structured counselling on safe sex practices and regular semen tests until virus RNA clearance has been verified. Viral persistence has been documented in Sudan virus disease, and in Ebola virus disease where the median time to negative PCR was 204 days and where risk-reduction in sexual practices has the potential to reduce transmission. The absence of such a programme could have catastrophic consequences in the re-emergence of infection and potential for a new outbreak.
3. Identification of visual loss known to occur at high rates as a result of uveitis following filovirus infection. Specialist treatment would be required, but the absence of a screening programme would necessarily mean patients did not receive any appropriate treatment (data not shown).

The risks of establishing this structured care environment were felt to be minimal, provided that significant costs (direct or indirect) were not incurred by patients or their families, and that stigma was reduced.

World Health Organization
Avenue Appia 20
CH-1211 Geneva 27
Switzerland